

SMART HOSPITAL ENERGY MANAGEMENT SYSTEM (SHEMS)

Production-Grade AI BEMS Architecture & Specifications Manual

AUTHOR:

Tanay Meshram (Full-Stack Energy BEMS Developer)

SYSTEM VERSION:

v1.0.0 (FastAPI Core + React SPA + SQLite/MySQL Engine)

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1. System Overview & Architecture

SHEMS is a production-grade, AI-powered building energy management system (BEMS) designed specifically for critical healthcare environments. It optimizes microgrid electricity routing, HVAC target temperatures, and standby medical equipment idle times in real time without compromising patient safety, clinical comfort, or critical life-support operations.

Hybrid Microgrid Routing Engine Hierarchy:

* Priority 1: Solar Power

Directly offsets hospital active load draw during sunlight hours. Excess solar generation is dynamically routed to charge the BESS battery cells.

* Priority 2: Battery Storage (BESS)

During peak load hours or high-tariff windows (2:00 PM - 6:00 PM), BESS batteries discharge to run auxiliary systems, shaving grid imports.

* Priority 3: Primary Utility Grid

Operated as the baseline grid importer to meet hospital draws when renewable solar and battery reserves are depleted.

* Priority 4: Emergency Generator Backup

During grid outages or critical battery drops (<20%), generator systems activate automatically to maintain safe conditions in the ICU and OT blocks.

2. Machine Learning Intelligence Pipelines

A. Demand Forecaster (Random Forest Regressor)

Learns hour-of-day, day-of-week, occupancy levels, and outdoor temperatures. It projects a continuous load forecast curve over the next 24 hours, letting the smart routing engine schedule pre-charging windows during low-tariff mornings.

B. Outlier Anomaly Detector (Isolation Forest)

Analyzes live grid import, carbon indices, and total draws. It detects statistical anomalies representing power spikes or line grounding faults. The autoencoder model also maps room-level draws to flag abnormal server draws (e.g. crypto-mining draft load in the Research Center).

C. Predictive Maintenance (Logistic Sigmoid Regression)

Computes asset State of Health (SOH) and Remaining Useful Life (RUL) using mechanical variables: compressor vibration (mm/s), core temperature (C), and oil pressure (PSI). If failure probability rises above 65%, a critical maintenance alert is automatically written to the database.

3. API Endpoints & MQTT Telemetry

Key REST API Endpoints:

*** GET /api/dashboard/live**

Returns current live JSON telemetry dict (occupancy, temperatures, powers).

*** POST /api/hvac/override**

Allows manual target overrides (room, lights, hvac mode, fan speed). Blocks overrides that violate ICU/OT patient safety shields.

*** GET /api/predictions/peak**

Returns predicted peak time, projected peak demand (kW), and load distributions.

*** GET /api/maintenance/predictive**

Exposes failure probabilities, RUL, and next service dates for Chillers, HVAC, Generators, UPS, and Inverters.

MQTT Broker Topics (HiveMQ Integration):

*** Topic: hospital/bems/telemetry/live**

Publishes the full JSON sensor telemetry dict every 3 seconds to keep UI visual graphs synchronized.

*** Topic: hospital/bems/alerts/live**

Publishes active critical warning messages when temperature drifts or power spikes occur.

4. Clinical Safety Guards & Automation

Clinical Safety Shield Locks (ICU & OT):

- Patient safety always has the highest priority. ICU and OT blocks are locked continuously to comfort climate settings.
- Remote state modifications (standbys or shut-offs) are completely blocked for all critical life-support machines (Ventilators, Cardiac Monitors, Infusion Pumps, Emergency Lights).
- HVAC target temp setpoints in critical zones are restricted within tight clinical margins (ICU: 20C - 23C, OT: 18C - 22C). All out-of-bracket requests are rejected at the FastAPI core.

Occupancy-Aware Automation Rules:

In non-critical wings (like Outpatient Clinic, General Wards, Admin offices), if occupancy count drops to 0 for 10 minutes (simulated at 30 seconds for quick validation):

- Lights & Fans are switched to OFF automatically.
- Displays & PC monitors are powered down to Standby.
- HVAC Target shifts to ECO Mode (24.5C), cutting HVAC load draw by up to 40%.
- Non-essential auxiliary draws are disabled.

As soon as a new occupant enters the zone, BEMS immediately wakes up the systems and restores Comfort Mode.

5. Schema Specifications & Local Installation

SQLAlchemy Database Schema Overview:

SHEMS uses MySQL 8.0 (with a local zero-dependency SQLite fallback) mapped through SQLAlchemy:

- * users: Admin, Energy Manager, and HVAC Tech accounts.
- * settings: Toggles for BESS shaving, occupancy audits, and HVAC target variables.
- * energy_readings: Log table tracking grid imports, solar generation, and carbon metrics.
- * equipment: List of critical, semi-critical, and non-critical assets with runtime and utilization tracking.

Local Installation Guide:

1. Start Backend FastAPI:

```
$ cd smart-hospital-energy
$ python -m venv venv
$ venv\Scripts\activate    # On Windows
$ pip install -r backend/requirements.txt
$ python backend/app.py    # Launches on http://localhost:5000
```

2. Start Frontend React:

```
$ cd frontend
$ npm install
$ npm run build            # Packages assets to backend/dist
$ npm run dev              # Launches hot-reload on http://localhost:5173
```